



VTA-27

For Every Memory That Matters

A First-in-Class Oral Therapeutic for
Alzheimer's Disease (AD)

August 2026



Disclaimer

This presentation is for informational purposes only and does not constitute an offer, commitment, or binding agreement. It contains forward-looking statements that involve risks and uncertainties, and actual results may differ materially. No representation or warranty, express or implied, is made regarding the accuracy, completeness, or reliability of the information provided. This document may include unpublished data and proprietary insights. It is intended solely for the recipient and should not be disclosed, distributed, or used for any purpose other than evaluating a potential partnership. VASTHERA assumes no liability for reliance on this information.

01

The Unmet Need

- Standard-of-Care
- Anti-Amyloid- β Therapies
- Neuroinflammation
- A Need for a Multi-target Approach

AD Treatments: SOCs (Symptomatic)

Feature	Aricept [Donepezil] (Eisai/Pfizer)	Exelon [Rivastigmine] (Novartis)	Razadyne [Galantamine] (Janssen)	Namenda [Memantine] (Abbvie)	Namzaric
	AChE Inhibitor			NMDA	Combination
Modality	Small molecule Acetylcholinesterase Inhibitor (AChEI)	Small molecule dual Cholinesterase Inhibition (AChE & BuChE)	Small molecule AChE inhibitor	Small molecule NMDA antagonist	Aricept + Namenda
Formulation	Oral tablet, ODT, Transdermal patch	Oral capsule, oral solution, Transdermal patch	Oral tablet, ER capsule, Oral solution	Oral tablet, ER capsule, Oral solution	Oral ER capsule
Target	Acetylcholinesterase (AChE)	AChE + Butyrylcholinesterase (BuChE)	AChE, nicotinic receptor modulation	NMDA receptor	AChE + NMDA Receptor
Safety	Nausea, Diarrhea, Insomnia, Bradycardia	Nausea, Vomiting, Diarrhea, Dizziness, Headache	GI toxicity, Dizziness	Dizziness, Confusion, Headache, Hallucinations	Dizziness, GI effects
Disease Stage	Mild-to-moderate	Mild-to-moderate	Mild-to-moderate	Moderate-to-severe	Moderate-to-severe

AD Treatments: Anti-Amyloid β Antibodies

Feature	Anti-amyloid β
Key assets	• Leqembi [Lecanemab] (Eisai/Biogen) • Kisunla [Donanemab] (Eli Lilly) • Trotinemab (Roche/Genentech) • Remternetug (Eli Lilly)
Target	Amyloid β
Stage	Validated
MoA	Clears toxic amyloid β species
Strength	• Robust clinical evidence • Clear biomarker-clinical outcome linkage • Commercial infrastructure rapidly scaling
Disadvantages	• ARIA risk • Modest efficacy • Does not fully stop the disease



KEY MESSAGE

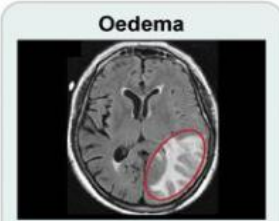
Amyloid removal alone does not address the biological drivers of neurodegeneration.



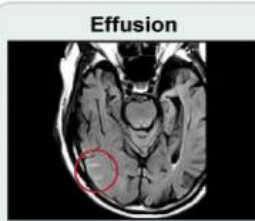
LIMITATIONS OF ANTI-AMYLOID β ANTIBODIES

1

AMYLOID-RELATED IMAGING ABNORMALITIES (ARIA)



Parenchymal oedema in the left parieto-occipital lobe (T₂-FLAIR) (solid circle)¹



Leptomeningeal effusion in several sulci within the right temporo-occipital lobe (T₂-FLAIR) (solid circle)¹



GRE, T₂* (solid circle)²



Small leptomeningeal haemosiderin deposit in the left frontal lobe sulcus (solid circle)¹

ARIA-E : Edema (Fluid accumulation or fluid leakage)

- Removal of plaque weakens blood vessels, inducing immune response and disrupt BBB

ARIA-H : Hemorrhage (Cerebral microhemorrhages and hemosiderin deposits)

- Proposed mechanism as BBB disruption and immune-inflammatory response

ARIA risk is highest in vulnerable patients, including APOE ϵ 4 carriers and those with vascular amyloid pathology, requiring MRI monitoring and careful patient selection.

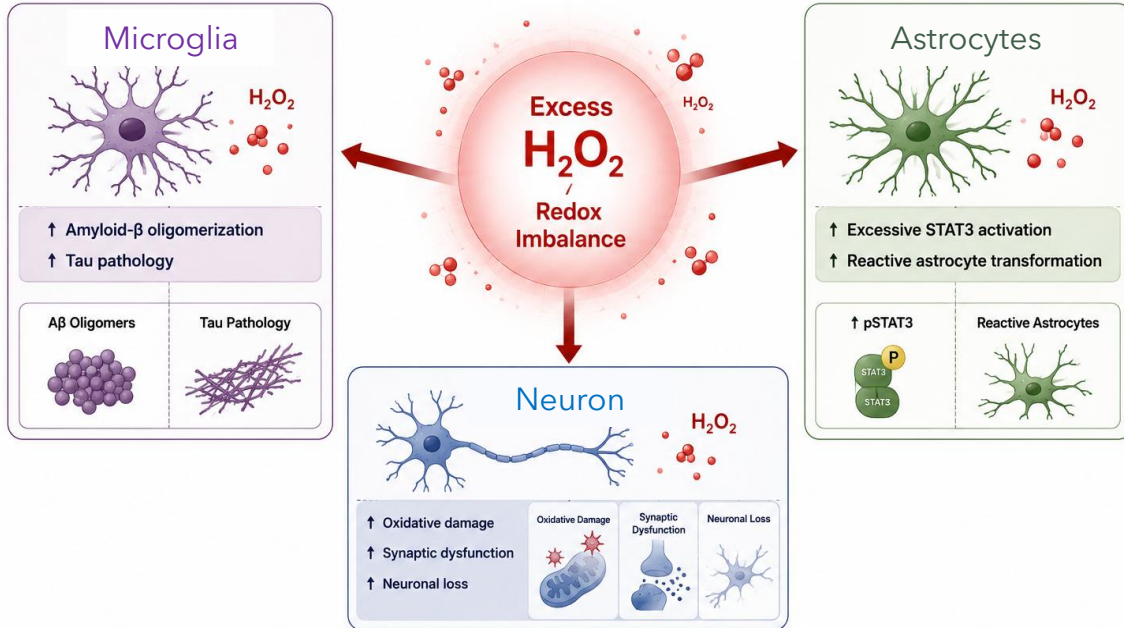
2

AMYLOID REMOVAL DOES NOT RESTORE COGNITIVE FUNCTION

Anti-amyloid antibodies reduce amyloid burden and modestly slow memory decline, but they do not restore lost cognition or reverse established neurodegeneration driven by synaptic dysfunction, tau pathology, chronic neuroinflammation, and neuronal loss. Amyloid removal alone is, therefore, insufficient for true disease modification.

Neuroinflammation: A Critical Link Between Disease Severity and Cognitive Loss

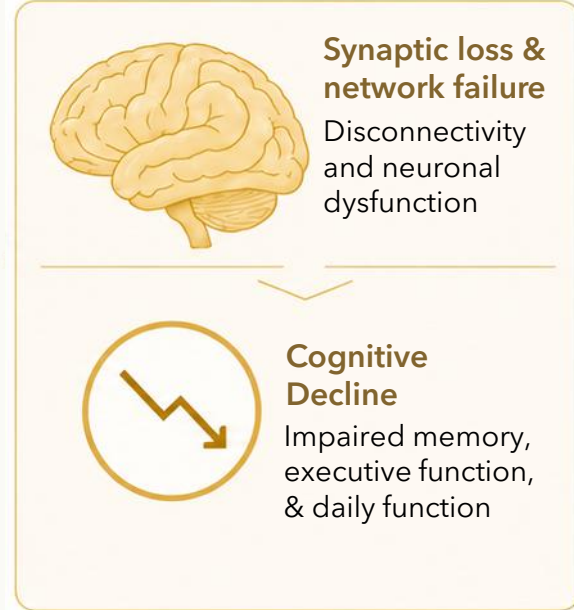
1 REDOX BIOLOGY IN GLIAL CELLS



2 FROM PATHOLOGY

	Amyloid Plaques Drive neurotoxicity and inflammation
	Tau pathology Disrupts microtubules and transport
	Neuroinflammation Sustained activation of innate immune response
	Neuronal Stress Energetic and proteostatic stress in neurons
	BBB Dysfunction Leads to leakage and immune cell infiltration

3 LEADING TO COGNITIVE DECLINE



4 EMERGING EVIDENCE IMPLICATES GLIAL PATHWAYS

GLIAL ASSOCIATED TARGETS

-  **APOE:** Lipid transport and astrocyte function
-  **TREM2:** Microglial activation and survival
-  **TAM (Tyro3/AXL/MER TK):** Phagocytosis and resolution of inflammation
-  **Astrocytic STAT3:** Astrocyte reactivity and neuroprotection

TRANSLATIONAL SUPPORT

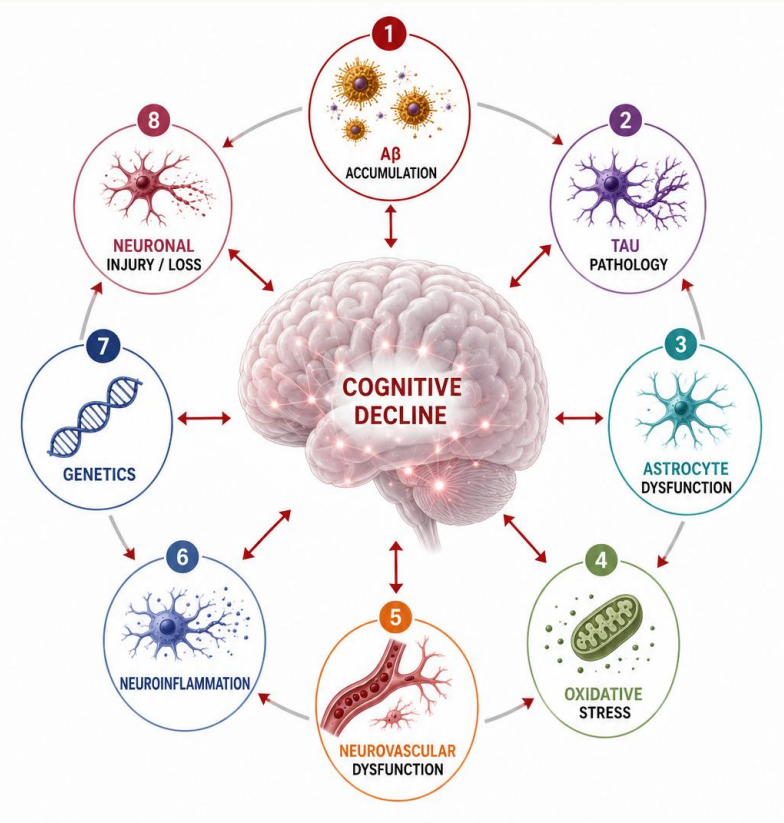
- ✓ Glial activation correlates with disease progression and cognitive decline
- ✓ Glial resilience associates with slower decline and resilience
- ✓ Preclinical models show that restoring glial function improves synaptic health and cognition

✓ KEY MESSAGE

Glial cell-targeted therapies go beyond amyloid clearance by restoring the neuro-immune axis essential for neuronal network and cognitive function.

Multi-target Approach is a solution for Disease Modification in Alzheimer's Disease

1 A NEED FOR A MULTI-FACETED APPROACH



2 CURRENT THERAPIES ADDRESS SELECTED COMPONENTS OF AD BIOLOGY

	 Neuron Protection Increases neuronal density	 Cognitive Recovery Improves learning, memory and function	 Anti-Neuroinflammation Restores chronic neuroinflammation	 Vascular Restoration Restores cerebral vascular health	 Amyloid Plaque Removal Reduces amyloid plaques
 Leqembi / Kisunla (anti-Aβ)					
 BIIB080 (anti-tau)					
 TREM2 / NLRP3					
 Neurotrophic (BDNF, NGF)					

✓ There is a need for a multi-target therapeutic that uniquely elicits neuronal restoration, cognitive recovery, anti-inflammatory activity, and vascular regeneration - addressing key downstream drivers beyond amyloid to support meaningful clinical benefit.

02

Brain Specific Mechanism of Peroxiredoxin

- Redox Signaling Mechanism
- Discovery and MoA of Peroxiredoxin
- In Vivo Validation
- In Vitro Validation



H₂O₂-mediated Redox signaling

- H₂O₂ is a second messenger in the membrane receptor-mediated signal transduction and in turn mediates intracellular signaling through protein oxidation, which modulates protein activity or interaction

Dr. Toren Finkel's Lab (1995)

Science

Requirement for Generation of H₂O₂ for Platelet-Derived Growth Factor Signal Transduction

Science 270 (5234), 296-299
DOI: 10.1126/science.270.5234.296



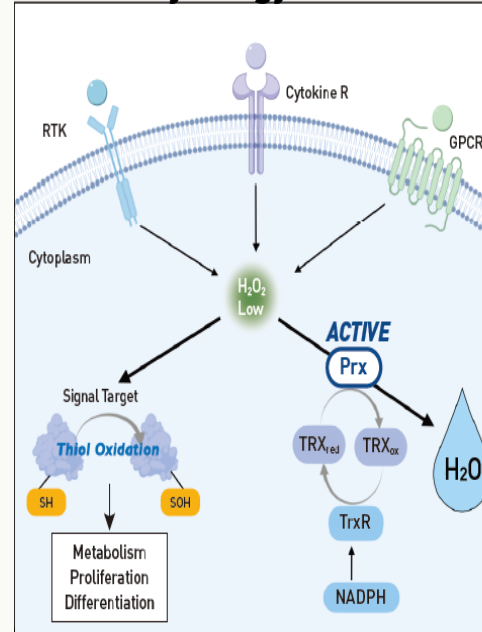
Dr. Sang Won Kang's Lab (2005)

nature LETTERS

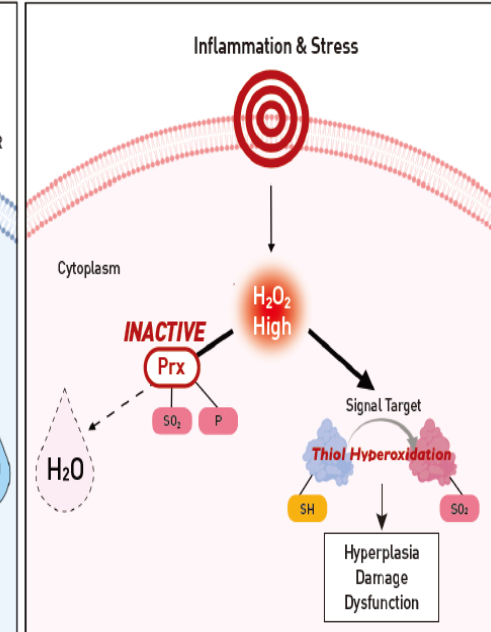
Regulation of PDGF signaling and vascular remodeling by peroxiredoxin II

Nature volume 435, pages347-353 (2005)
DOI: 10.1038/nature03587

Normal Physiology



Diseased State



Wood, Poole, Karplus (2003)

Science

Peroxiredoxin Evolution and the Regulation of Hydrogen Peroxide Signaling

Science 300 (5619), 650-653
DOI: 10.1126/science.108040



Woo et al. (2003)

Science

Reversing the Inactivation of Peroxiredoxins Caused by Cysteine Sulfinic Acid Formation

Science 300 (5619), 653-656
DOI: 10.1126/science.1080273

Peroxiredoxin Functions as a Redox signaling Modulator

Dr. Earl Stadtman's Lab (1988)

JBC | JOURNAL OF BIOLOGICAL CHEMISTRY

The Isolation and Purification of a Specific "Protector" Protein Which Inhibits Enzyme Inactivation by a Thiol/Fe(III)/O₂ Mixed-function Oxidation System*

J Biol Chem. 1988 Apr 5;263(10):4704-11.
DOI: 10.1016/S0021-9258(18)68840-4



Dr. Sue Goo Rhee's Lab (1994, 1998)

JBC | JOURNAL OF BIOLOGICAL CHEMISTRY

Thioredoxin-dependent Peroxide Reductase from Yeast* (1994)

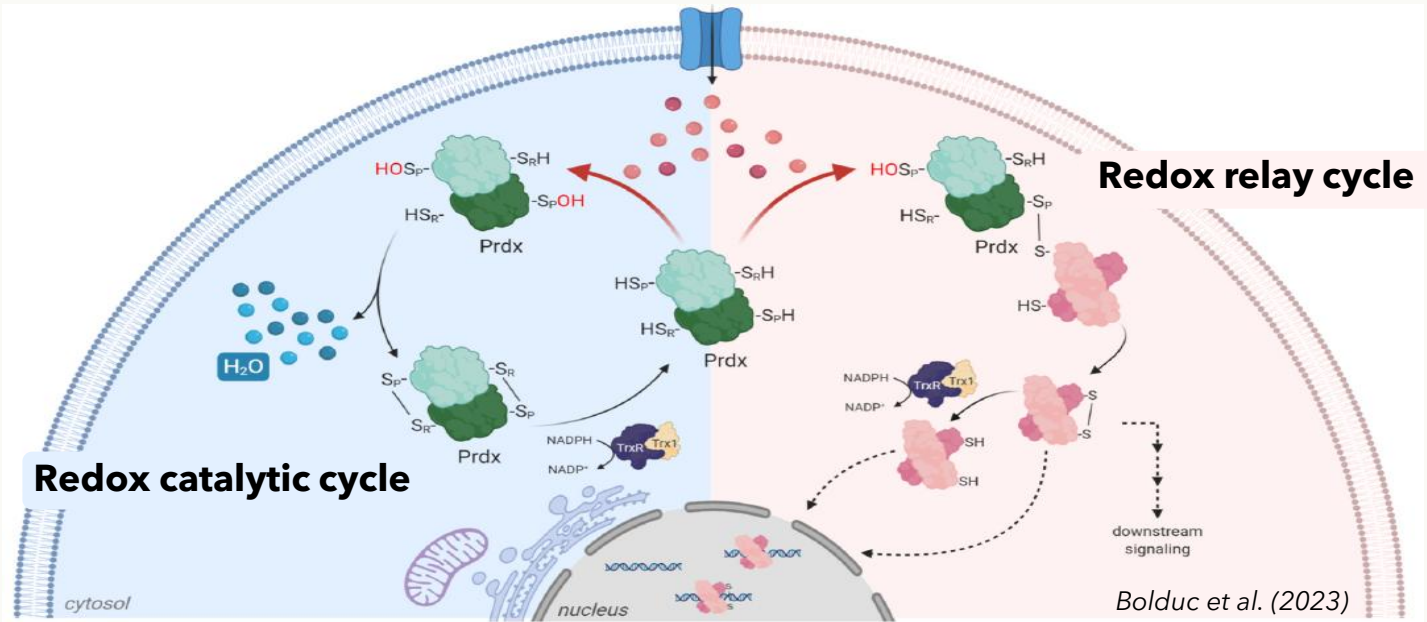
J Biol Chem. 1994 Nov 4;269(44):27670-8.
DOI:10.1016/s0021-9258(18)47038-x

Mammalian Peroxiredoxin Isoforms Can Reduce Hydrogen Peroxide Generated in Response to Growth Factors and Tumor Necrosis Factor- α * (1998)

J Biol Chem . 1998 Mar 13;273(11):6297-302.
DOI: 10.1074/jbc.273.11.6297.

Sang Won Kang[§], Ho Zoon Chae[§], Min Seok Seo[§], Kanghwa Kim[§], Ivan C. Baines^{**} ^{§§}, and Sue Goo Rhee[§] ^{§§}

From the [§]Laboratory of Cell Signaling and ^{**}Laboratory of Cell Biology, NHLBI, National Institutes of Health, Bethesda, Maryland 20892



Dual Functions of PRX as a Redox Signaling Modulator

1) Signal Fidelity

PRX reacts with H₂O₂ with **high specificity** and sensitivity, maintaining its physiological level. This control is essential for preserving the fidelity of signaling pathways that rely on H₂O₂ as a **2nd messenger** in their feedback regulation. **Loss of functional PRX leads to signaling dysregulation through persistent loop activation.**

2) Signal Modulation

PRX also transfers oxidative equivalents to select proteins with redox-sensitive motifs through a process known as **Redox Relay**. This reversible oxidation modulates the activity and functional state of target proteins, allowing **precise fine-tuning of signal transduction.**

Novel target: Peroxiredoxin inactivation exacerbates AD and cognitive decline

1 BIOLOGICAL RATIONALE

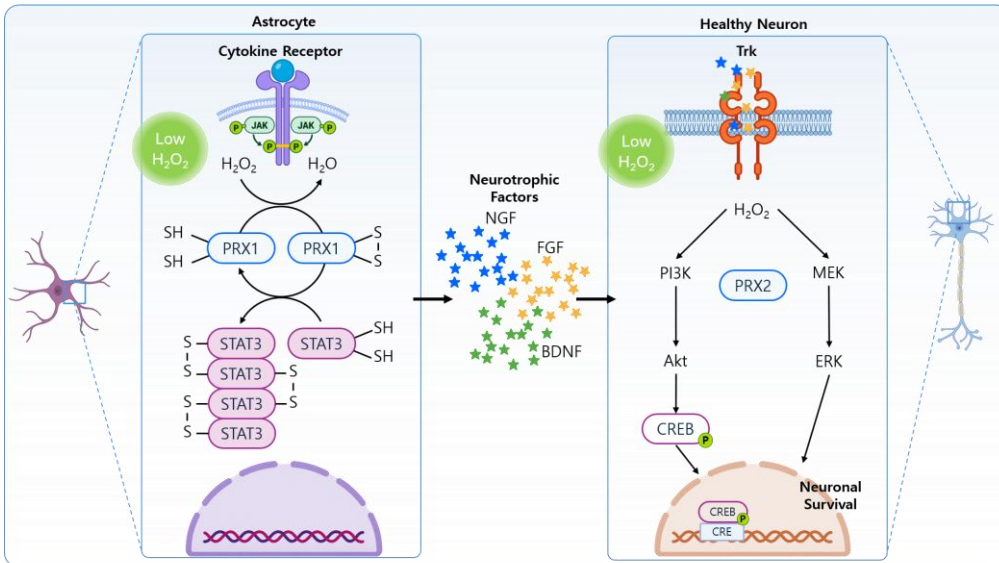
Peroxiredoxin: The Guardian of Redox Signaling

(PRX, gene name: PRDX)

Peroxiredoxins are thiol peroxidase enzymes that regulate cellular H_2O_2 as a catalytic substrate. They are essential for all aerobic organisms by tuning cellular redox signaling as well as further protecting cells from oxidative stress. They are the most sensitive enzyme against cellular H_2O_2 , thus described as the "Redox guardian".

CELL-TYPE SPECIFIC FUNCTION OF PRX1 AND PRX2 IN THE BRAIN

In the brain, PRX1 isoform is abundantly expressed in astrocytes, while PRX2 isoform is expressed selectively in neurons. Thus, both isoforms are believed to be involved in the astrocyte-neuron communication network essential for memory and cognitive functions.



Nature chem biol Sobotta MC et al, 2014; Frontiers in Immunol, Zhu G et al, 2022; Neurobiol Aging, Kim SU, 2011

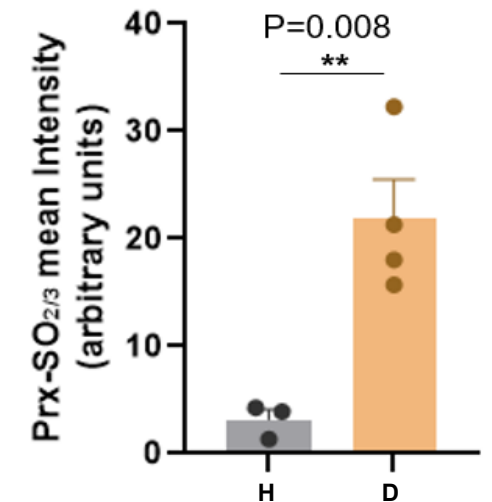
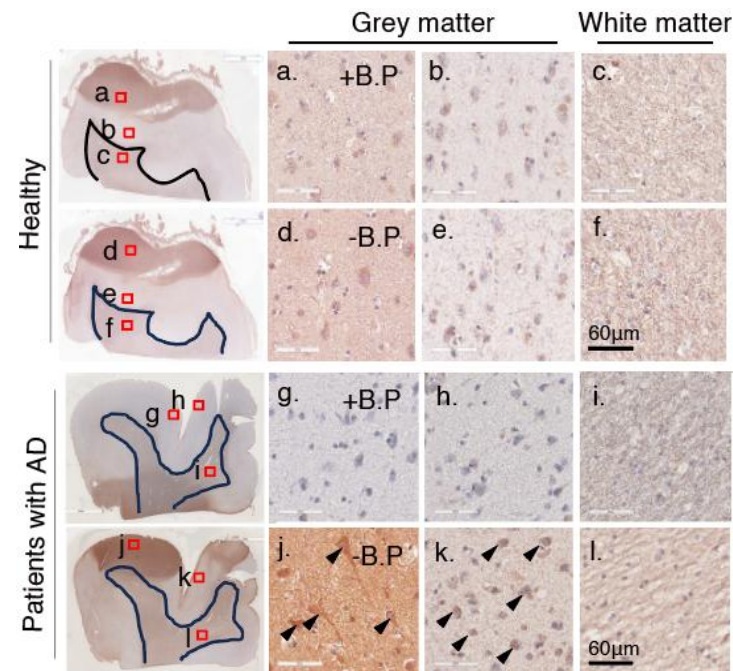
2 IMMUNOSTAINING FOR HYPEROXIDIZED PRX (DEAD ENZYME FORM)

STUDY DESIGN

Post-mortem brain tissues from AD patients (n = 4) and age-matched healthy controls (n = 3) were stained for hyperoxidized PRX (PRX-SO₂/₃) - the catalytically inactive form of the enzyme - by immunohistochemistry. Grey and white matter were assessed separately, antibody specificity was confirmed with a blocking-peptide control (+B.P / -B.P), and grey-matter signal intensity was quantified and compared between groups.

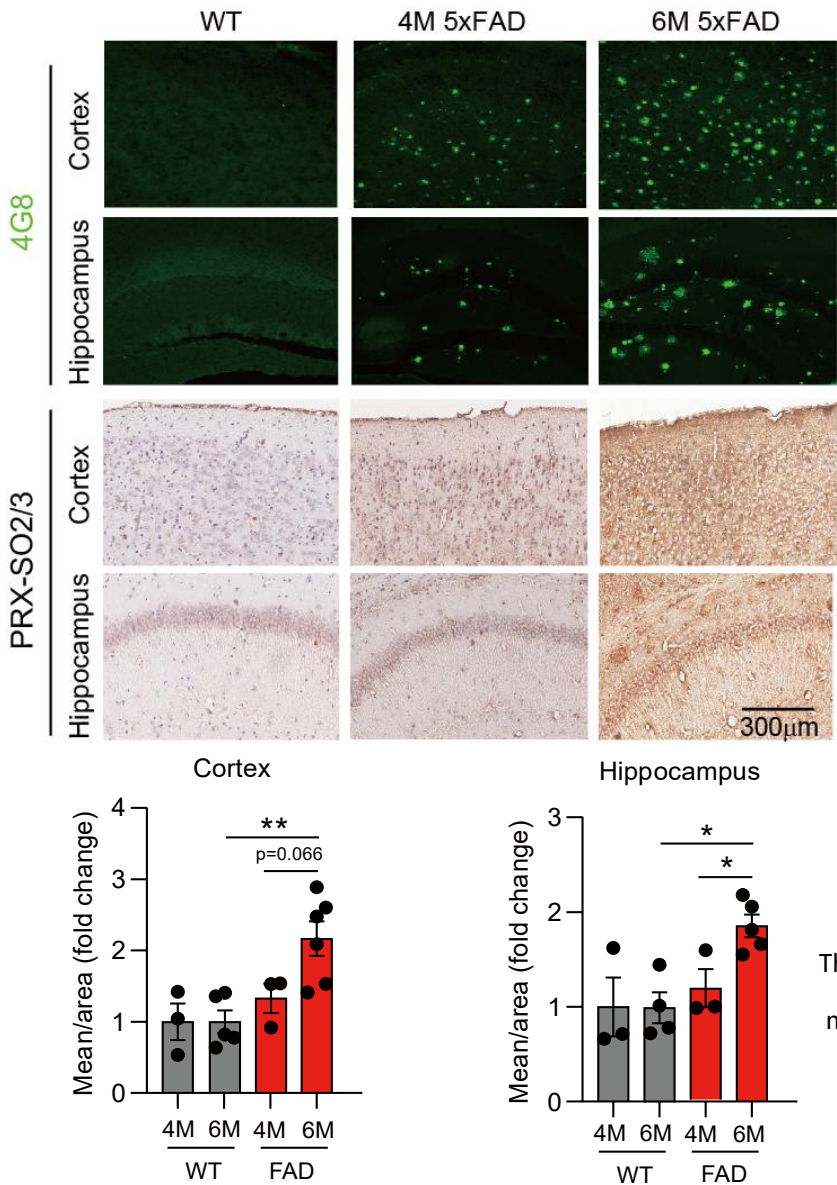
STUDY RESULTS

Compared to healthy brains, patients with AD showed increased level of PRX hyperoxidation. This suggests a possibility that PRX inactivation may exacerbate the pathogenesis of AD.



Key: H = Healthy / D = Patients with AD

Peroxiredoxin inactivation in the Brain of AD Mouse Model



STUDY DESIGN

To determine whether the PRX hyperoxidation seen in human AD is recapitulated in vivo, the same analysis was conducted in wild-type and 5xFAD mice at 4 and 6 months of age. Using the same PRX-SO_{2/3}-specific antibody, hyperoxidized PRX (PRX-SO_{2/3}) - the dead enzyme form - was detected by immunohistochemistry in the cortex and hippocampus. Signal was quantified as fold change in mean intensity/area relative to WT. Antibody specificity was confirmed by a blocking peptide control.

STUDY RESULTS

The level of hyperoxidized PRX (PRX-SO_{2/3}) is significantly increased in correlation with disease progression in the 5xFAD brain.

Dead PRX levels were slightly detectable at 4 months and rose significantly by 6 months in both the cortex and hippocampus. This demonstrates that PRX inactivation is not a consequence of the disease, but a progressive, disease-associated event conserved across human AD and 5xFAD mouse models.

KEY MESSAGE

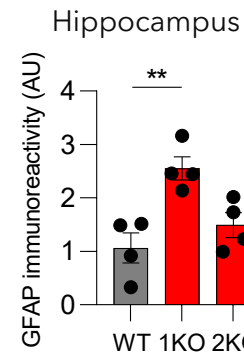
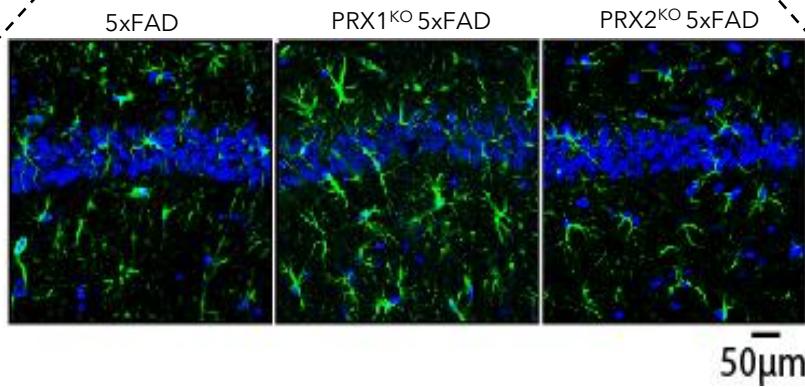
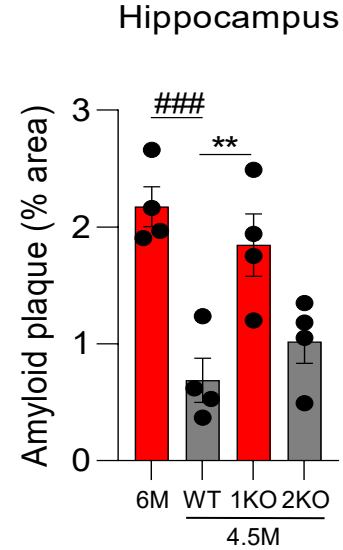
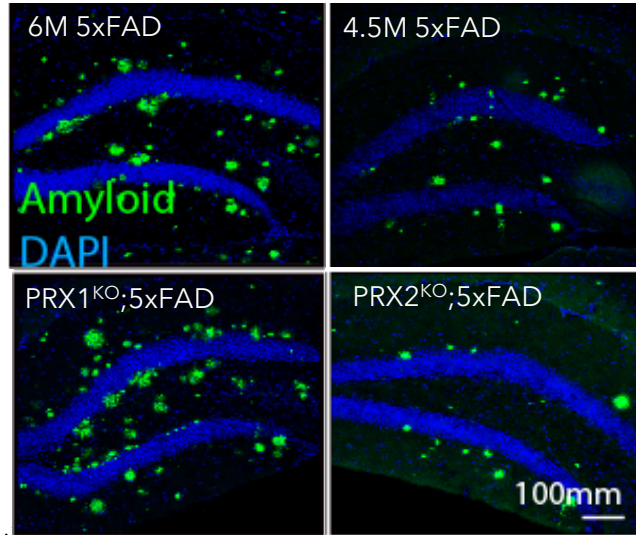
PRX inactivation accumulates in step with disease progression and is conserved across human and mouse models, strengthening the rationale for targeting PRX in AD.

PRX1/2 Deficiency Accelerates A β Plaque Formation



All groups: 5xFAD background

Hippocampus



Key: 6M = 6 month 5xFAD / WT = 4.5 month 5xFAD
1KO = 4.5 month PRX1 KO: 5xFAD / 2KO = 4.5 month PRX2 KO: 5xFAD

STUDY DESIGN

To test whether astrocytic PRX1 controls amyloid pathology in vivo, 5xFAD mice were crossed with PRX1 or PRX2 knockout lines and analyzed at 4.5 months of age - an early time point before plaque load plateaus. Representative hippocampal sections were co-stained for amyloid plaques (4G8) and the reactive-astrocyte marker GFAP, and signal was quantified as plaque area (% area) and GFAP immunoreactivity (AU) across 5xFAD, 5xFAD;PRX1KO and 5xFAD;PRX2KO groups, with 6-month 5xFAD included as an advanced-pathology reference.

STUDY RESULTS

- At 4.5 months, plaque burden was still low in 5xFAD mice relative to the 6-month reference. Against this background, PRX1 deletion significantly increased hippocampal amyloid plaque load, reaching levels comparable to 6-month 5xFAD mice, whereas PRX2 deletion did not.
- PRX1 loss also drove a parallel rise in GFAP immunoreactivity, indicating that the same early time point shows both accelerated amyloid deposition and reactive astrogliosis in PRX1-deficient mice.

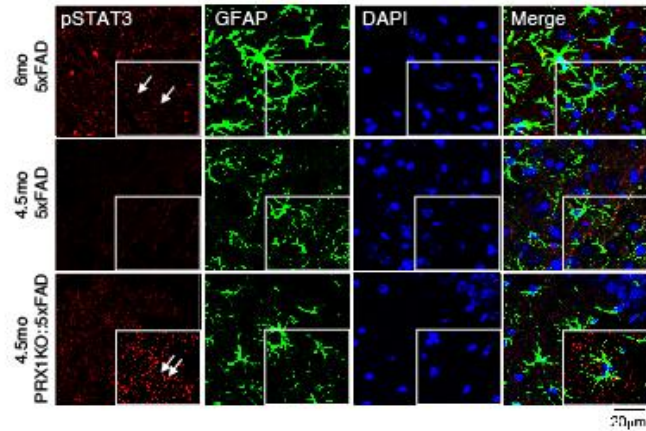


KEY MESSAGE

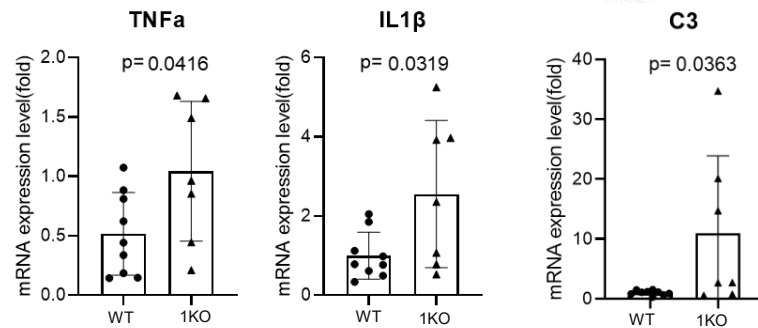
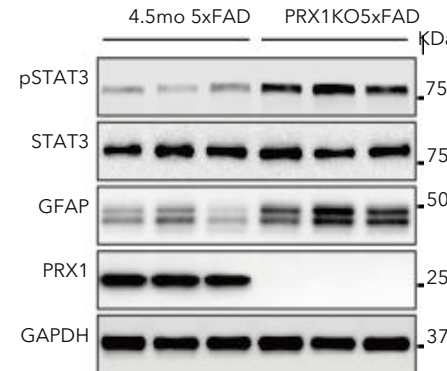
Astrocytic PRX1, not neuronal PRX2, gates early amyloid pathology. Its loss accelerates plaque deposition and reactive astrogliosis in young 5xFAD mice.

PRX1 Deficiency Exacerbates Astrocyte Reactivation Through STAT3 control

IN VIVO MOUSE MODEL



Cortex



Key: WT = 5xFAD / 1KO = PRX1KO5xFAD

STUDY DESIGN

PRX1-knockout mice were crossed with 5xFAD mice and analyzed at 4.5 months. Reactive astrogliosis and STAT3 activation were assessed by GFAP/pSTAT3 staining and Western blot, with cortical cytokine mRNA measured by qPCR.

STUDY RESULTS

PRX1 deficiency increased the number of reactive astrocytes.

PRX1 deficiency gave rise to increases in pSTAT3 levels in both cortex and hippocampus - major AD-pathology regions accompanied by elevated pro-inflammatory cytokines (TNFα, IL-1β, and C3).



KEY MESSAGE

(In vivo) PRX1 deficiency → STAT3 hyperactivation → astrocyte reactivation.

PRX1 is a direct, redox-operated brake on STAT3; its loss in AD releases this brake – a novel, targetable mechanism.

STAT3 Is the Master Regulator of Astrocyte Reactivity

- **STAT3 plays a distinct role in each CNS cell type**

Microglia

Context-dependent inflammatory modulator

Microglial cells respond to not only anti-inflammatory cytokine (IL-10), but also pro-inflammatory cytokines (IL-6/TNFα) inducing microglial hyperactivation.

Keren-Shaul et al., Cell, 2017, Krasemann et al., Immunity, 2017

Astrocyte

Key driver of reactive astrogliosis

STAT3 activation drives GFAP induction, hypertrophy, inflammatory mediator expression, and maintenance of reactive astrogliosis.

Neuron

Developmental lineage regulator

STAT3 acts mainly in development, gating the neurogenic-to-gliogenic switch and progenitor expansion. It does not drive an adult reactive disease state.

Pollen et al., Cell, 2015, Taga & Fukuda, Clin Rev Allergy Immunol, 2005

Astrocyte STAT3 dependency is validated across five independent lines of evidence

NECESSITY

Loss of function

Astrocyte-specific STAT3 deletion blocks GFAP induction, hypertrophy and scar formation

Herrmann 2008; Okada 2006

SUFFICIENCY

Gain of function

Activating JAK2-STAT3 alone induces astrocyte reactivity without a pathological stimulus

Ceyzeriat 2018

OMICS-SCALE

Transcriptome & chromatin

STAT3 reprograms hundreds-thousands of reactive DEGs and chromatin accessibility as a core regulator

Burda 2022; Wei 2021

AD RESCUE

Disease-hallmark impact

Astrocyte STAT3 deletion lowers plaque burden and inflammation and rescues cognition in AD mice

Reichenbach 2019; Ben Haim 2015

HUMAN

Translational relevance

TNF-STAT3-SERPINA3 axis drives reactive astrocytes in human iPSC / BBB models

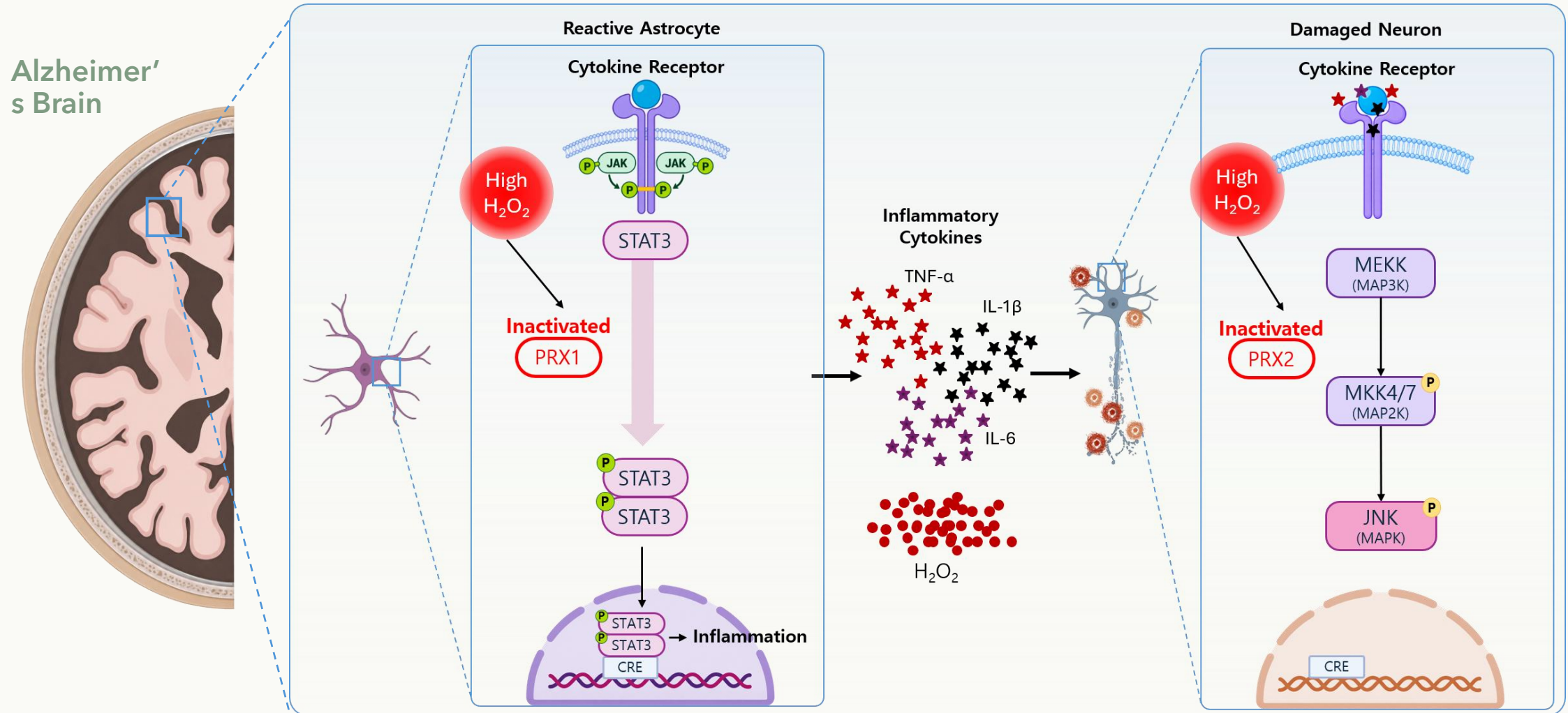
Kim 2022



KEY MESSAGE

STAT3 is a key driver of reactive astrogliosis critical for onset of neuroinflammation.

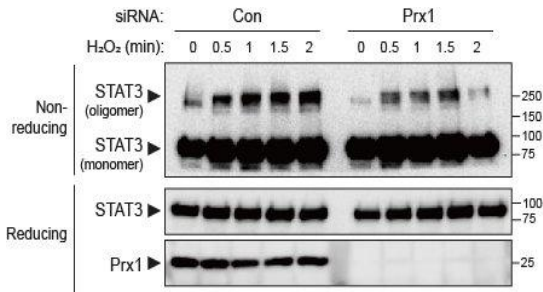
MoA: Brain-Specific PRX-Cored Neuroinflammation Circuit in AD



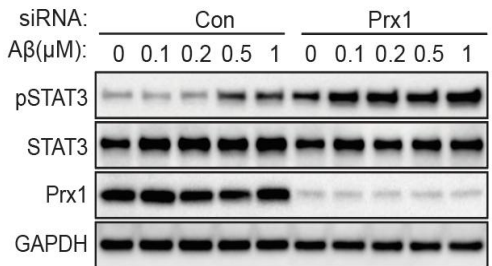
PRX1 Inhibits STAT3 Activation and Reduces Cytokine Production

PRIMARY ASTROCYTES

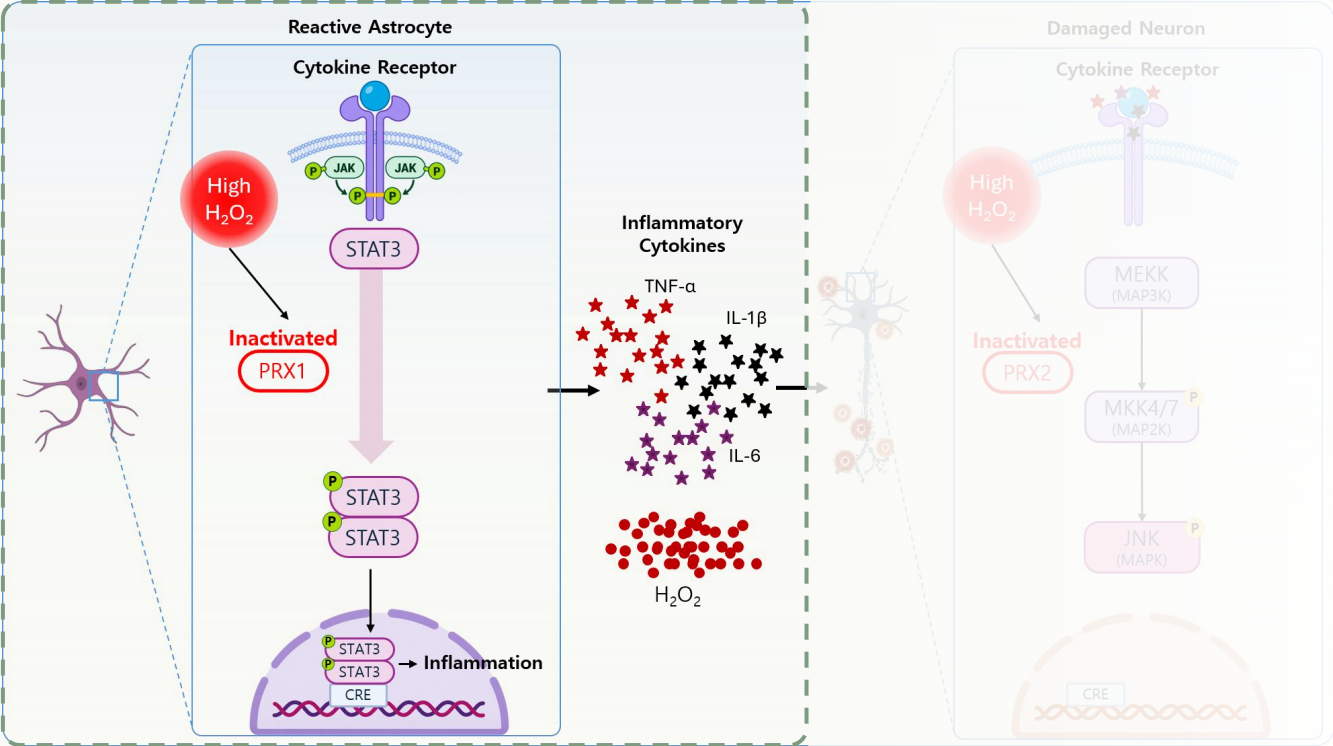
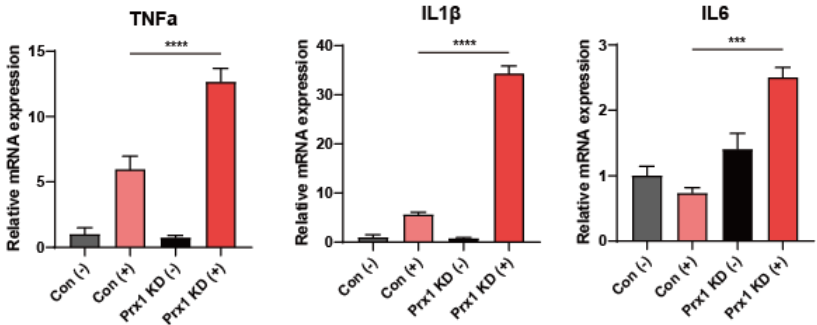
Reduced STAT3 oligomerization by PRX1 absence



Increased STAT3 phosphorylation by PRX1 absence



Increased STAT3 > Increased cytokines



STUDY DESIGN

In primary astrocytes, PRX1 was depleted by siRNA to measure STAT3 oligomerization, phosphorylation, and cytokine production (Western blot). SBP pull-down tested direct, H₂O₂-dependent PRX1-STAT3 interaction.

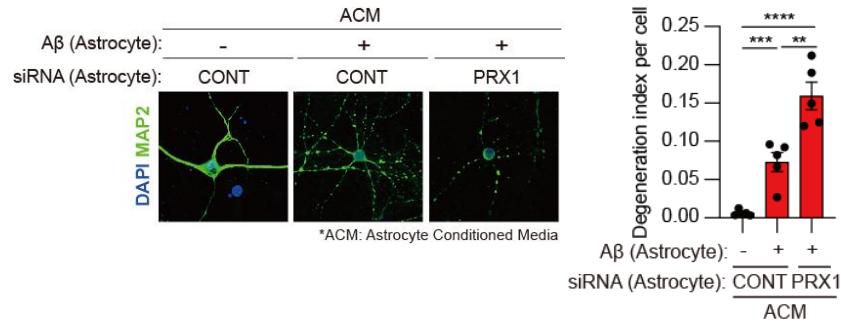
STUDY RESULT

PRX1 senses H₂O₂ and relays it to STAT3 by direct redox interaction, promoting STAT3 oligomerization and thereby restraining its activation.

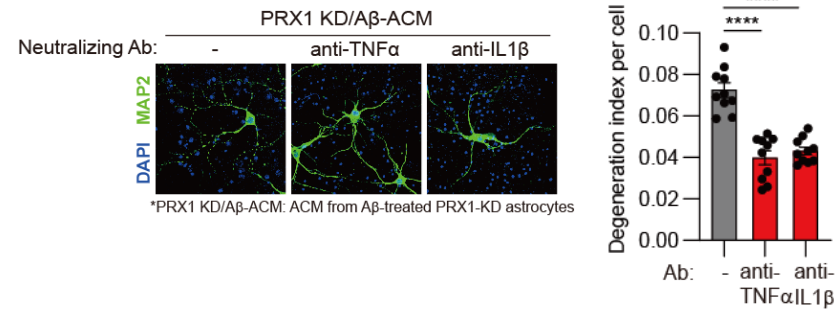
PRX2 Inhibits Inflammatory Cytokine-Induced Neuronal Death

PRIMARY NEURONS

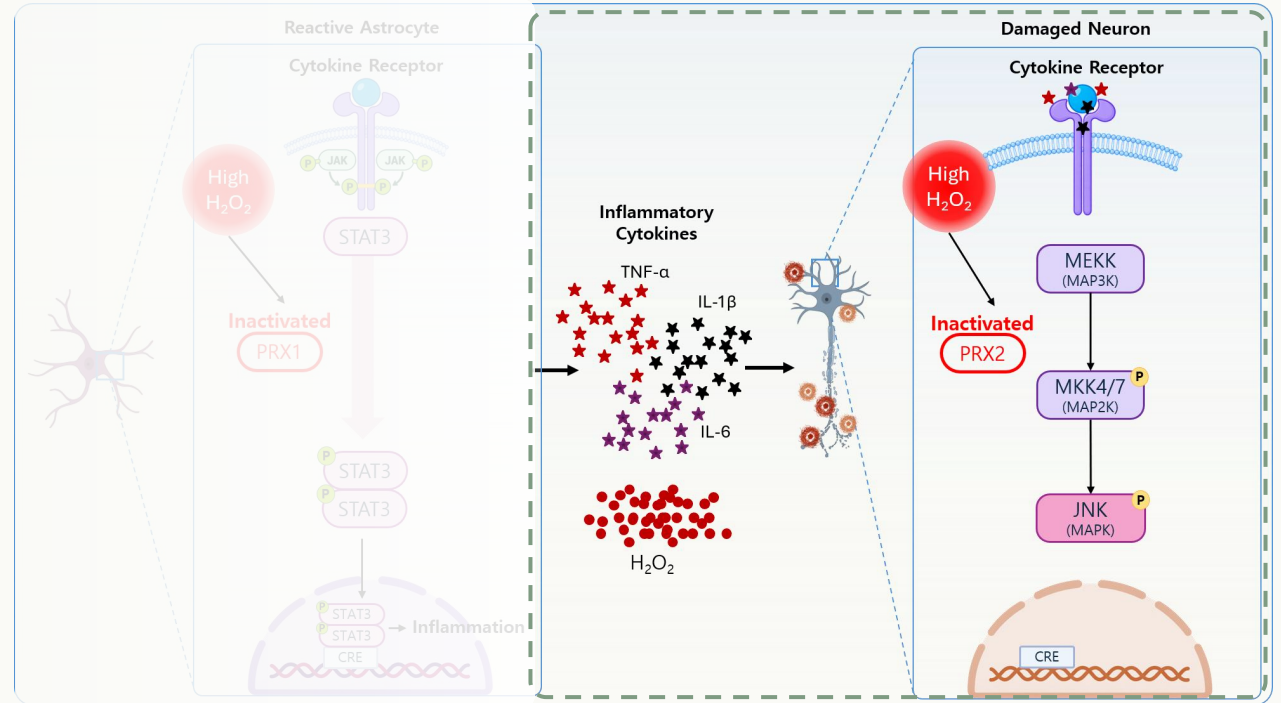
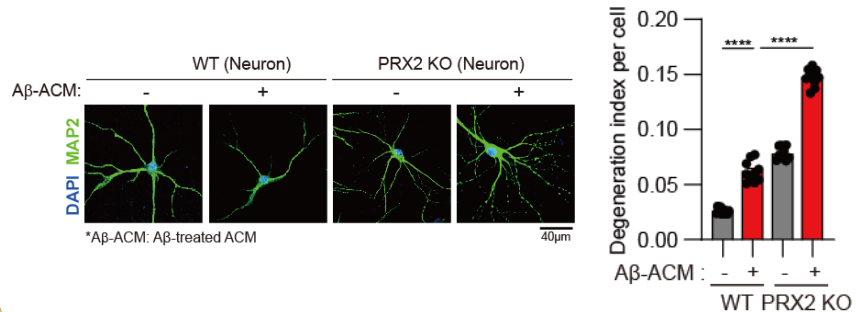
Reactive astrocyte-derived factors induce neuronal death



Reactive astrocytes produce pro-inflammatory cytokines



PRX2 protects neurons against astrocyte-derived cytokines



STUDY DESIGN

- Primary astrocytes were transfected with control or PRX1 siRNA and treated with A β oligomers. The conditioned media (A β -ACM) was collected and treated mouse primary neurons, and neurite degeneration was quantified by MAP2 immunostaining.
- The A β -ACM was preincubated with TNF α - and IL-1 β -neutralizing antibodies to quench the reactive astrocyte-derived cytokines.

STUDY RESULT

Astrocytes depleted of PRX1 convert to reactive state and produce pro-inflammatory cytokines, which trigger neuronal death. The neuronal PRX2 plays a protective role from reactive astrocyte-derived pro-inflammatory cytokines. **This establishes a PRX-cored neuroinflammation circuit.**

03

VTA-27

- In vitro activity of VTA-27
- In vivo efficacy of VTA-27
- Mode of Action of VTA-27
- Competitive Landscape



VTA-27 Precisely Mimics the Enzymatic Function of PRX in vitro

VTA-27 (PRX mimetic)

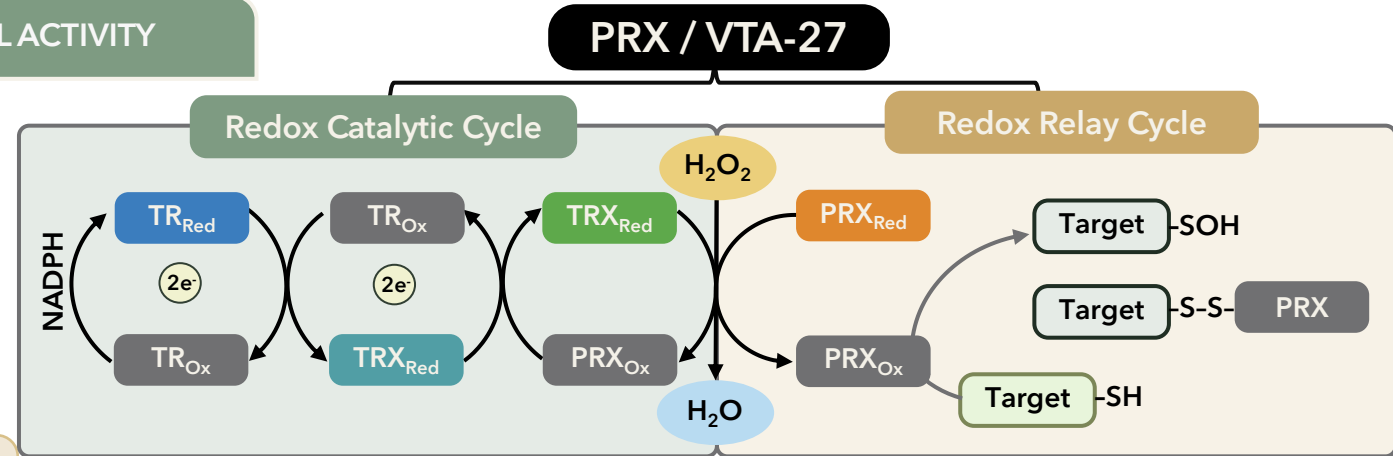
- A small molecule that mimics PRX catalytic and redox relay activities
- Compensates for inactivated PRX in AD
- As a PRX mimetic, VTA-27 is a new modality called **Protein Oxidation Modulator (POM)**

STUDY DESIGN

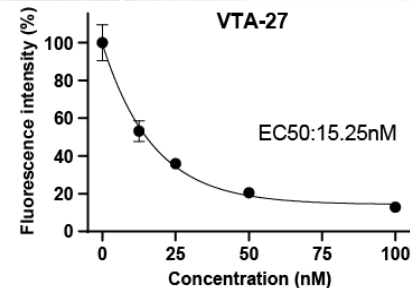
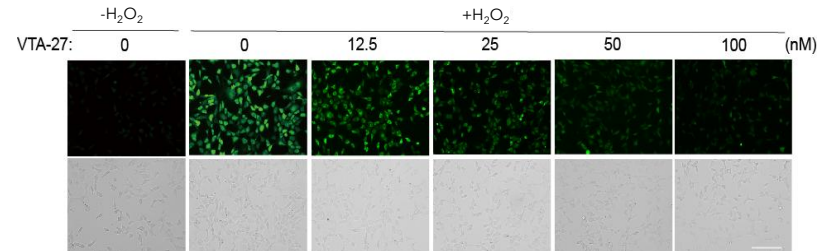
This study was conducted to measure catalytic and redox relay activities of VTA-27 as a PRX-mimetic small molecule.

(Cells: NIH3T3)

PRX/VTA-27 DUAL ACTIVITY

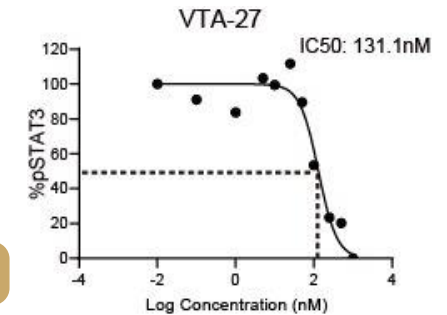


In Vitro RESULTS



Catalytic Cycle

Relay Cycle



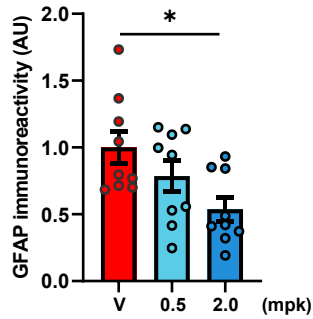
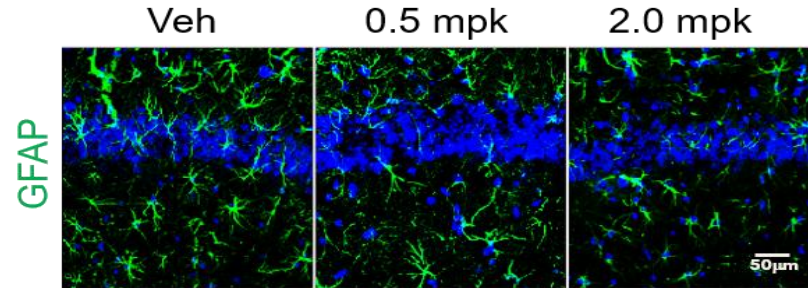
KEY MESSAGE

VTA-27 eliminates intracellular H_2O_2 levels by its peroxidase activity and simultaneously inhibits the $A\beta$ -induced STAT3 activation by its redox relay activity.

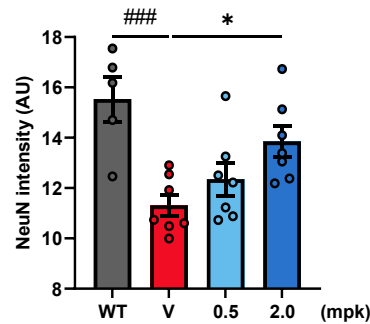
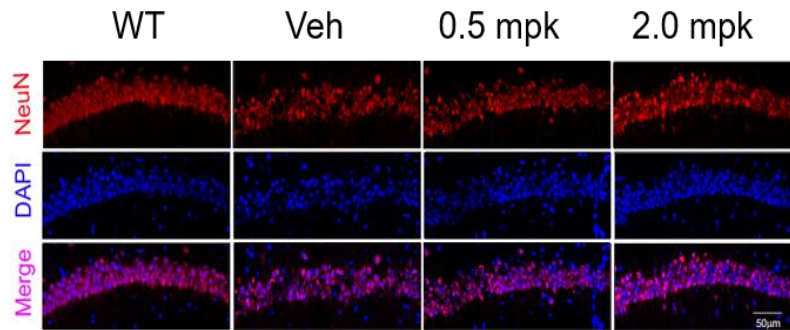
VTA-27 inhibits pathological features of AD

VTA-27 Inhibits Astrocyte Activation

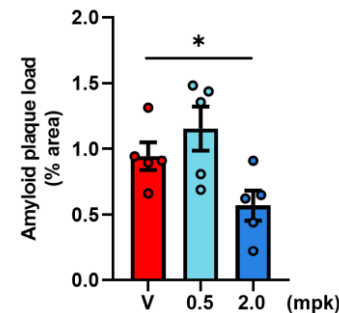
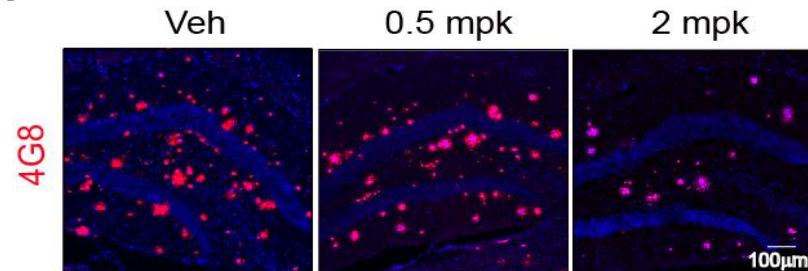
Astrocyte (GFAP)



Neuron (NeuN)



Aβ burden (4G8)



STUDY DESIGN



4.5-month-old 5xFAD mice were treated with VTA-27 (0.5 or 2mg/kg, i.p., twice weekly) for 8 weeks. Following behavioral assessment, brain tissues were collected for immunofluorescence analysis of GFAP, Aβ (4G8), and NeuN in AD-relevant brain regions.

STUDY RESULTS

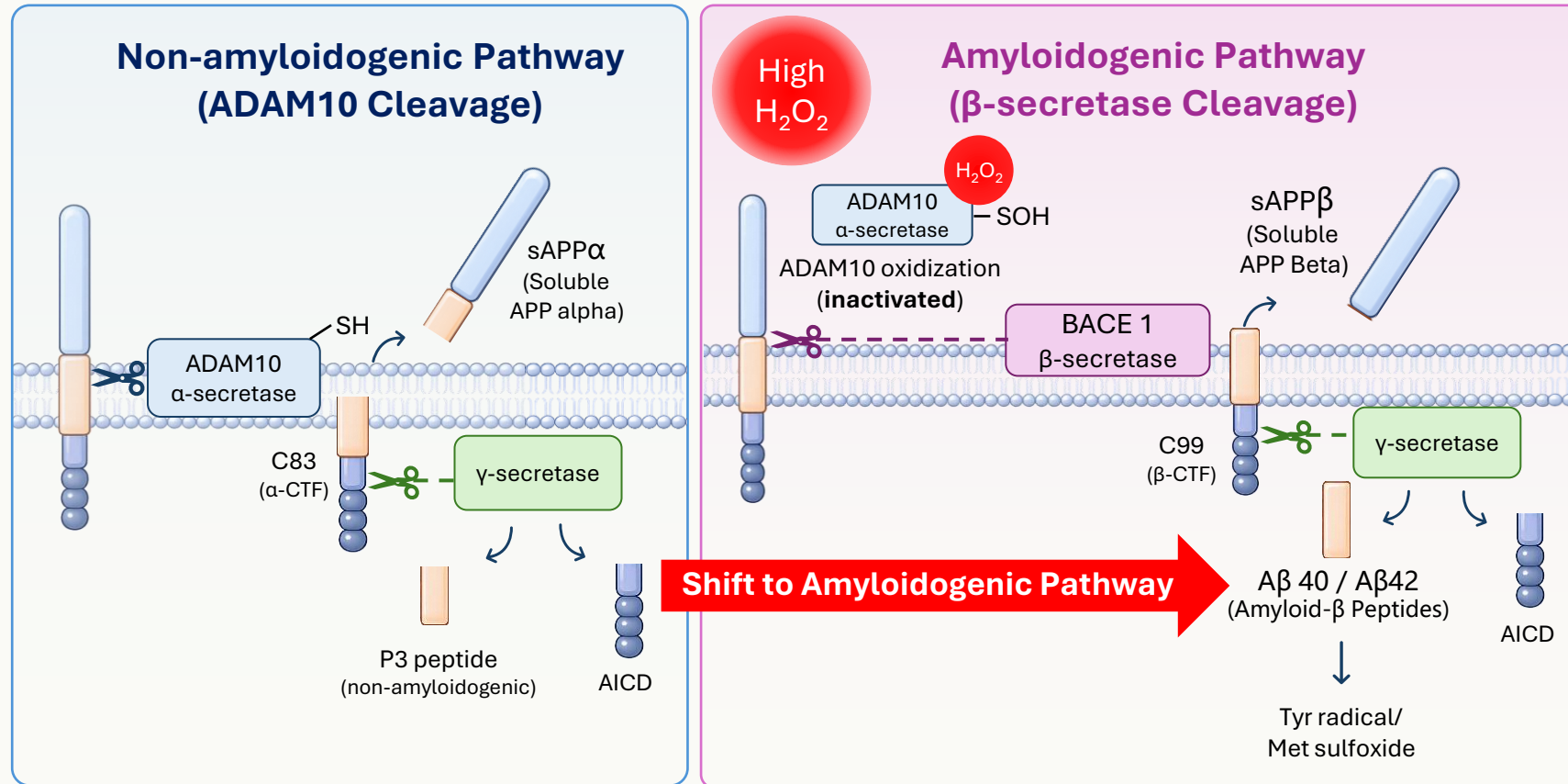
- Representative hippocampal IF analysis showed that VTA-27 modulated AD-related neuropathology in 5xFAD mice.
- At 0.5 mg/kg, VTA-27 treatment showed early biological signals, including a trend toward reduced GFAP-positive reactive astrogliosis and increased NeuN-positive neuronal density, although Aβ plaque burden was not clearly reduced.
- At 2 mg/kg, VTA-27 treatment produced broader pathological improvements, reducing both GFAP-positive astrogliosis and Aβ plaque burden while preserving NeuN-positive neurons.

✓ KEY MESSAGE

VTA-27 shows a dose-dependent improvement in hippocampal AD pathology, such as reducing reactive astrogliosis and preserving neuronal integrity in 5xFAD mice. In addition, the Aβ burden was somehow reduced at a high dose

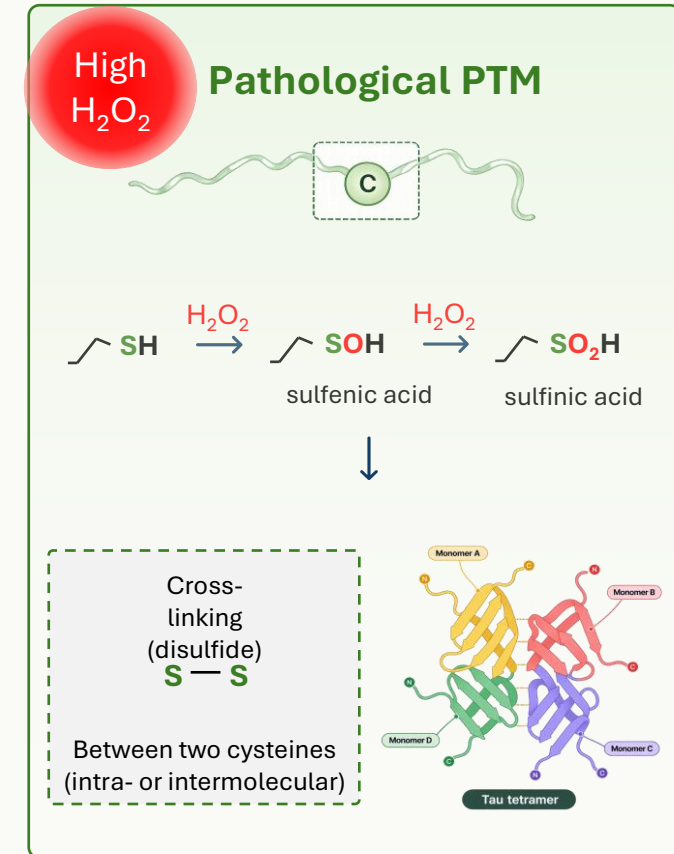
H₂O₂ Triggers Amyloid and Tau Aggregation Through Protein Oxidation

Amyloid Beta (Aβ) Production



ADAM10 vs BACE1: Two Competing Pathways that Determine Aβ Production

Tau Aggregation



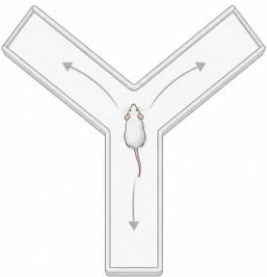
POTENTIAL MECHANISM

Excess H₂O₂ activates amyloidogenic pathway through the oxidation of ADAM10 driving A-beta peptide production.

Excess H₂O₂ triggers Tau oligomerization through cysteine oxidation.

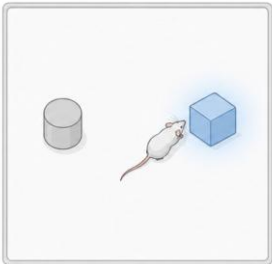
VTA-27 Restores Cognitive Impairment

Y-MAZE



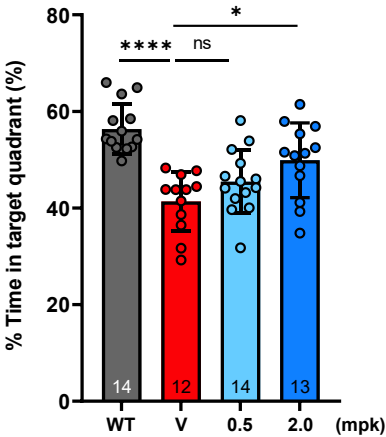
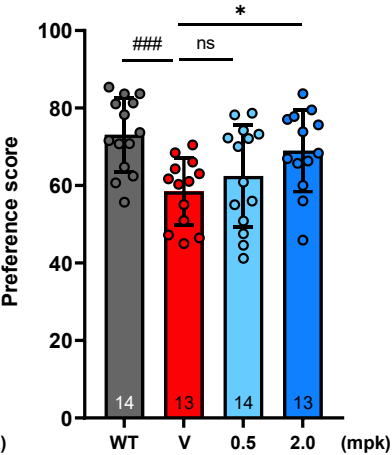
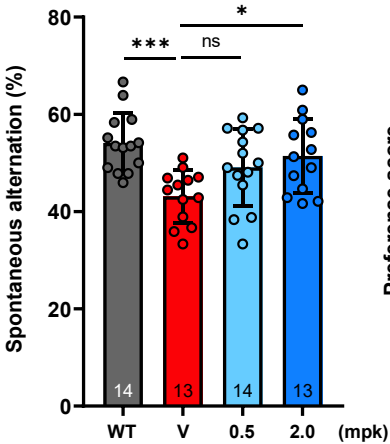
NOR

Novel Object Recognition



MWM

Morris Water Maze



STUDY DESIGN



Following 8 weeks of VTA-27 treatment, cognition-related behavioral performance was assessed using Y-maze, Novel Object Recognition, and Morris Water Maze tests, covering working memory, recognition memory, and spatial learning/memory.

STUDY RESULTS

Vehicle-treated 5x FAD mice (V) showed impaired performance across cognition-related behavioral assays. VTA-27 treatment showed a recovery trend at 0.5 mg/kg and more consistent improvement at 2 mg/kg across Y-maze, NOR, and MWM, supporting functional benefits across multiple memory domains.



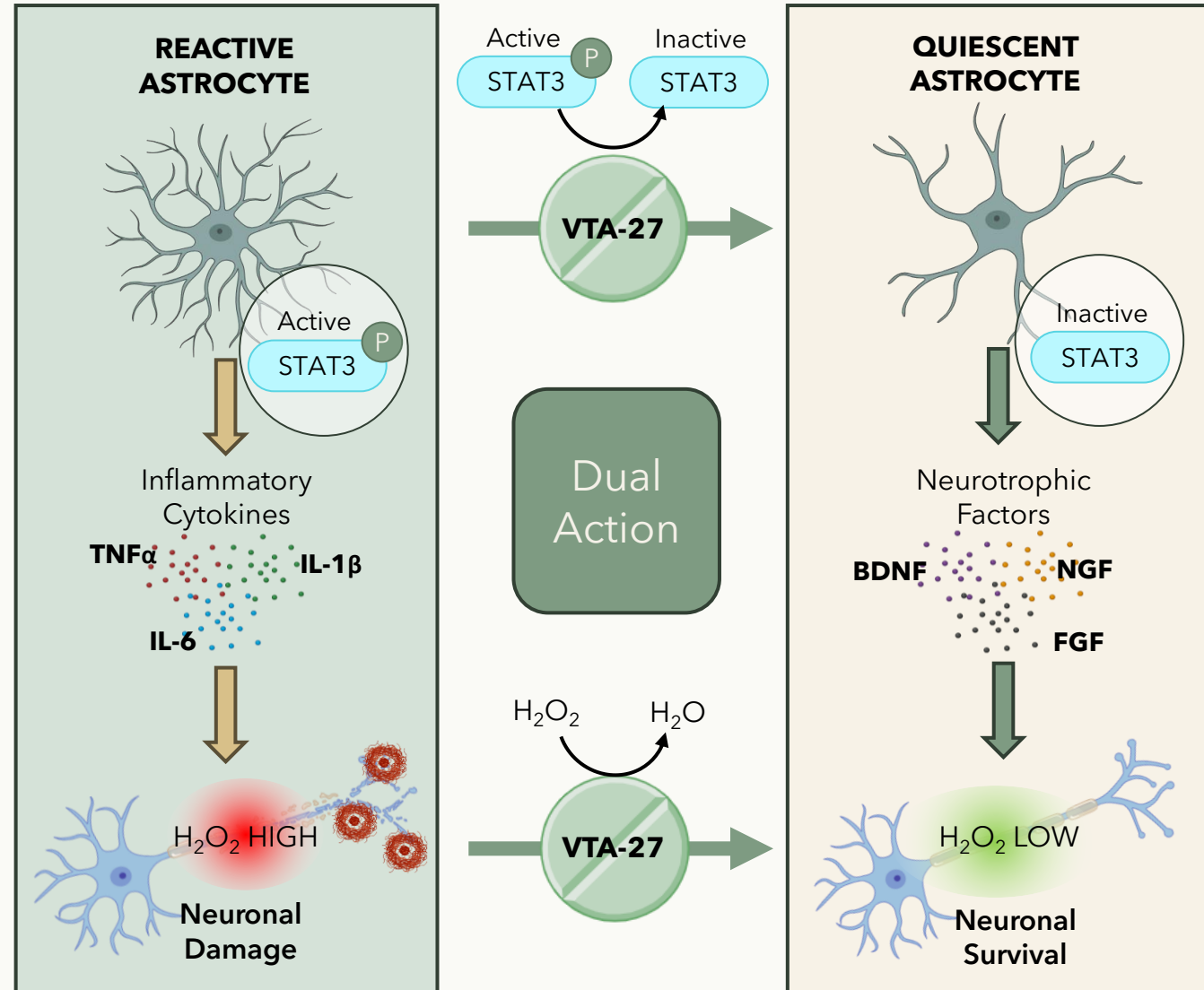
KEY MESSAGE

VTA-27 improves cognition-related behavioral performance in 5x FAD mice, supporting functional translation of its hippocampal neuroprotective effects.

VTA-27: A First-in-Class Neuroinflammation Targeting Therapy

✓ KEY MESSAGE

VTA-27 is one of the protein oxidation modulator (POM) and designed to restore cognitive function by simultaneously shutting down reactive astrocytes and protecting neurons, offering a differentiated neuroinflammation targeting approach for Alzheimer's disease.



Beyond Amyloid: VTA-27 Addresses the Key Downstream Drivers of Alzheimer’s Disease

	 Neuron Protection Increases neuronal density	 Cognitive Recovery Improves learning, memory and function	 Anti-Neuroinflammation Restores chronic neuroinflammation	 Vascular Restoration Restores cerebral vascular health	 Amyloid Plaque Removal Reduces amyloid plaques
 Leqembi / Kinsula (anti-A)					
 BIIB080 (Tau ASO)					
 TREM2 / NLRP3					
 Neurotrophic (BDNF, NGF)					
 VTA-27					



VTA-27 can be a multi-target therapeutic by uniquely eliciting neuronal restoration, cognitive recovery, anti-inflammatory activity, and vascular regeneration - addressing key downstream drivers beyond amyloid to support meaningful clinical benefit.

04

IND-Enabling Data

- Candidate Profile
- Current Development Status
- Development Timeline



Candidate Profile

Items	Contents
Indication	Alzheimer's Disease (AD)
Target Population	Mild to moderate AD patients
Substance	Small-molecule compound (VTA-27)
Administration route and frequency	Oral administration, once daily
MoA	VTA-27 inhibits astrocyte reactivation through redox-modulation of STAT3 activity
In vitro efficacy	Inhibition of STAT3 phosphorylation in primary astrocytes (IC ₅₀ = 162nM)
In vivo efficacy	Short working memory improvement at 2mg /kg dose (i.p. twice weekly for 8 weeks) in 5xFAD mouse Verified by external CRO: Y maze, Novel object recognition, Morris water maze
DMPK	ADME: in vivo PK (mouse, rat, dog & monkey), rat distribution and rat excretion as well as in vitro ADME are underway
Toxicity	GLP Tox study: 28-day repeat-dose toxicology studies completed in both rat and dog models. Genotoxicity & Safety Pharmacology: Ames test is negative. Safety pharmacology battery is clear, including hERG (IC ₅₀ >200 µM).
CMC	Synthetic route optimized with high purity (>99%) and low total impurities (<0.5%). - DS (GMP) production is underway - DP (GMP, active & placebo tablets) manufacturing is planned
Intellectual property	Material IP (PCT) ready to file

Current Development Status

COMPLETED

- **Non-Clinical**
 - Pharmacology Data
 - Pharmacokinetics Data
 - Toxicology Data

ONGOING / PLANNED

- **Non-Clinical**
 - Toxicokinetics Data
- **CMC**
 - Manufacturing
 - DS / DP (including stability data)
- **Clinical**
 - PK Modeling
 - IB Development

PHASE 1 CLINICAL TRIAL IMPLEMENTATION

- **Phase 1 Clinical Trial (First-in-Human)**
 - MRSD
 - Design
 - Dose Escalation (SAD & MAD)
 - **Endpoints**
 - Safety / Tolerability
 - PK
 - PD (exploratory)
 - **Investigational Site and PI Selection**

REGULATORY DESIGNATIONS & IND STATUS

- CRO Selection
- Pre-IND Meeting (if necessary)
- MFDS or FDA IND submission

Beyond IND: Additional Research Initiatives

1 Mechanistic Deep-Dive in Patient-Relevant Systems

- Extend the PRX-STAT3 redox-relay mechanism from mouse models into graded human AD tissue and patient-relevant neuron / astrocyte systems.
- Correlate PRX-SO₂/₃ (inactivated PRX) with disease severity and confirm the mechanism at cell-type resolution.

→ Strengthens the human translational relevance of the target.

2 Independent in vivo Efficacy Confirmation

- Oral (PO) dosing of VTA-27 in the 5xFAD model, reproduced at an independent external CRO under blinded, third-party conditions.
- Confirms the core efficacy signals observed internally – amyloid plaque reduction, reactive-astrocyte suppression, and cognitive rescue.

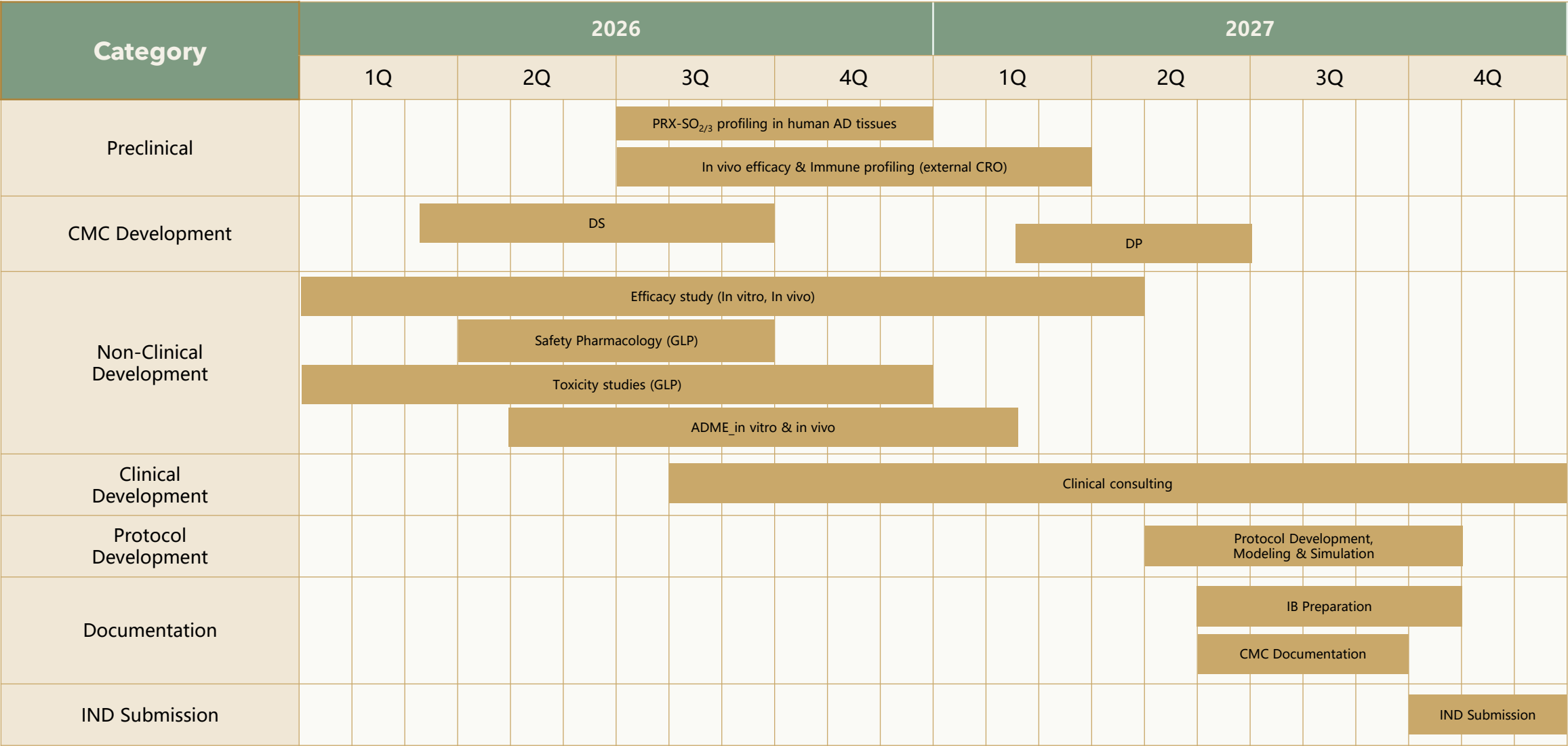
→ De-risks internal efficacy data for partners

3 Immune System Profiling

- Profile immune-cell subsets, cytokine responsiveness, IL-6-driven pSTAT3 signaling, and antigen-specific IgM / IgG generation in normal mice.
- Demonstrate preservation of physiological immune competence while selectively normalizing disease-associated astrocytic STAT3 hyperactivation.

→ Defines the target's selectivity window and supports a favorable therapeutic index.

Development Timeline



05

Company Overview

Company Profile

About VASTHERA

VASTHERA Co., Ltd. is a pharmaceutical company developing first-in-class small molecule drugs through its proprietary drug discovery platform Redoxizyme™ to treat cardiovascular, neurodegenerative, and oncologic disorders.

Our Mission

VASTHERA aims to develop innovative drugs that 'cure' diseases rather than simply 'alleviate' them, ultimately benefiting patients.

80%
R&D employees

40 ea
Global IPs

3 Assets
First-in-Class
Therapeutics

Location: Seoul, South Korea

Members: 17 employees / 3 SAB

Technology Platform: Redoxizyme™

Government & Public Grants

~12.6M USD Secured

KDRF

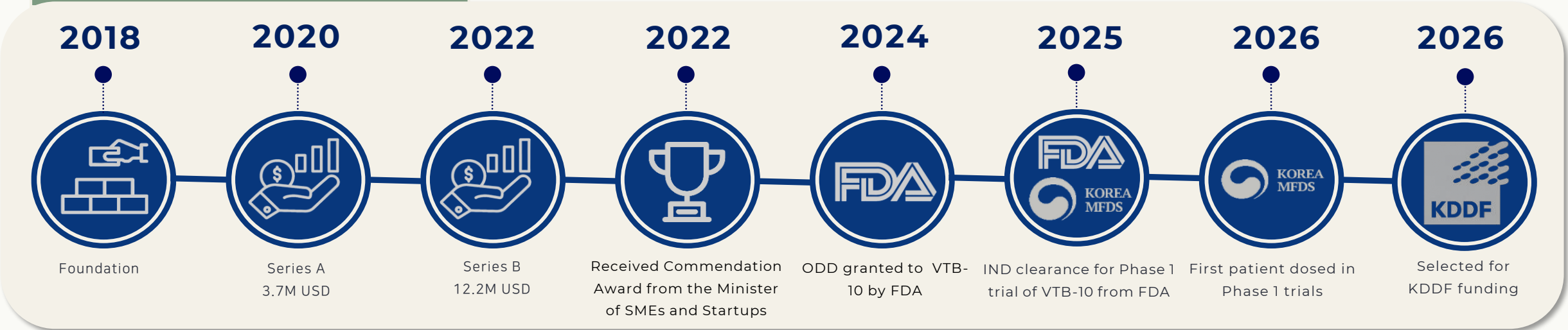
NRF 한국연구재단

KDDF 국가신약개발재단 Korea Drug Development Fund

koita 한국산업기술진흥협회 Korea Industrial Technology Association

KHIDI 한국보건산업진흥원

Company History



Our Pipeline

Pipeline	Indication	Discovery	Lead Optimization	Pre-Clinical	Phase 1	Note
VTB-10	 PAH					▪ ODD granted by FDA in 2024 ▪ IND filing in June 2025 ▪ Phase 1 started in 1Q of 2026 ▪ other CVD indications
VTA-27	 AD					▪ GLP-Tox started in 4Q of 2025 ▪ Potential expansion to PD
VTA-63	 TNBC					▪ GLP-Tox started in 4Q of 2025 ▪ Potential expansion to other cancers

Executive Leadership

VASTHERA's leadership team unites expert scientists and industry veterans with extensive experience in redox biology, medicinal chemistry, medical science, and cardiovascular research from leading institutions in the Republic of Korea and the US.



Sang Won Kang, PhD
CEO and Founder

Professor at Ewha Woman's University, and former CSO at LabFrontier, with careers in redox biology spanning academia, biotech, and NIH.



Kyung Joo Lee, PhD
CTO

20+ years in pharma and biotech, specializing in medicinal chemistry, with senior R&D roles in South Korea and the US, including Scripps Research Institute.



Hyun Gyu Lee, MD, PhD
CDO

10 years+ of experience as a venture capitalist specializing in Healthcare and Biotech, with senior leadership roles including VP and COO positions at multiple Biotech companies.



Dong Hoon Kang, PhD
CSO

Medical science expert, Research Professor at Ewha Woman's University and Asan Medical Center. Awarded the President's Postdoctoral Fellowship.



Choong Hee Lee, MSc
Director of Clinical Development

17+ years in global and local clinical trials across pharma, CROs, and biotech, with experience in Clinical Development and Operation for Phase 1-4 study, and for Medicines and Healthcare products.

For Every Memory That Matters

Contact Us



Phone
+82-2-3277-4908



Mail
bd@vasthera.com



Website
www.vasthera.com



Headquarters
#504, Science Building C,
52, Ewhayeodae-gil, Seodaemun-gu, Seoul, Korea



Research institute
#251, University-Industry Cooperation Building,
150, Bugahyeon-ro, Seodaemun-gu, Seoul, Korea